

Unit 1 - Quantum Mechanics

Quantum mechanics is a fundamental theory in physics that describes the physical properties of nature at the scale of atoms and subatomic particles.

1. Quantum mechanics is the foundation of several related disciplines including nanotechnology, condensed matter physics, quantum chemistry, structural biology, particle physics, and electronics.
2. The term “quantum” comes from the Latin word “quantus,” meaning “how much.” It refers to the discrete units of energy, charge, and other physical properties that are predicted by quantum theory.
3. Quantum mechanics was developed in the early 20th century to explain the behavior of matter and energy at the atomic and subatomic level.
4. Some of the key principles of quantum mechanics include wave-particle duality, uncertainty principle, and superposition.
5. Quantum mechanics has been extremely successful in explaining a wide range of phenomena, including the stability of atoms, the nature of chemical bonds, and the behavior of electrons in solids.
6. Despite its success, there are still many open questions and active areas of research in quantum mechanics, including the interpretation of the theory and its relationship to classical physics.

Inadequacy of classical mechanics for the notes of the Unit 1 - Quantum Mechanics in the subject of ENGINEERING PHYSICS

Classical mechanics is a branch of physics that explains the motion of celestial bodies such as planets, stars, and microscopic terrestrial bodies that move with non-relativistic speed. However, classical mechanics has its limitations and inadequacies, particularly in the region of atomic dimensions. Some of the inadequacies of classical mechanics are:

1. **Blackbody Radiation:** Classical mechanics could not explain the observed spectra of black body radiation .
2. **Photoelectric Effect:** Classical mechanics could not explain the photoelectric effect.
3. **Hydrogen Emission:** Classical mechanics could not explain the hydrogen emission of light.

These problems within classical physics led to the development of quantum theory. Quantum mechanics is a fundamental theory in physics that provides a description of the physical properties of nature at the scale of atoms and subatomic particles. It is the foundation of all quantum physics including quantum chemistry, quantum field theory, quantum technology, and quantum information science.

Planck's theory of black body radiation(qualitative)

- Planck's law describes the spectral density of electromagnetic radiation emitted by a black body in thermal equilibrium at a given temperature T , when there is no net flow of matter or energy between the body and its environment .
- Planck's radiation law is a mathematical relationship formulated in 1900 by German physicist Max Planck to explain the spectral-energy distribution of radiation emitted by a blackbody .
- A blackbody is a hypothetical body that completely absorbs all radiant energy falling upon it, reaches some equilibrium temperature, and then reemits that energy as quickly as it absorbs it .
- Using Planck's law, a black body's emitted radiation may be termed by its frequency variation. Planck's radiation, a form of thermal radiation, is another name for this type of radiation. The greater the temperature of the body, the more radiation of all wavelengths is emitted by it .
- The blackbody radiation problem was solved in 1900 by Max Planck. Planck used the same idea as the Rayleigh–Jeans model in the sense that he treated the electromagnetic waves between the walls inside the cavity classically, and assumed that the radiation is in equilibrium with the cavity walls .

Compton Effect

- Compton effect refers to the increase in the wavelength of photons (X-rays or gamma rays), due to their scattering by a charged particle (usually an electron).
- The impact has ended up being one of the foundations of quantum mechanics, which represents both wave and particle properties of radiation.
- The American physicist Arthur Holly Compton clarified (in 1922 and published in 1923) the frequency increment by considering X-rays as made out of discrete pulses, or quanta, of electromagnetic energy.
- Compton effect is the decrease in energy (increase in wavelength) of an X-ray or gamma ray photon, when it interacts with matter.
- Because of the change in photon energy, it is an inelastic scattering process.
- Inverse Compton scattering also exists, where the photon gains energy (decreasing in wavelength) upon interaction with matter.
- Compton effect, also called Compton scattering, is a principal way in which radiant energy is absorbed in matter.
- The Compton effect is the term used for an unusual result observed when X-rays are scattered on some materials.
- By classical theory, when an electromagnetic wave is scattered off atoms, the wavelength of the scattered radiation is expected to be the same as the wavelength of the incident radiation.

de-Broglie concept of matter waves

- According to the de Broglie concept of matter waves, the matter has dual nature. It means when the matter is moving it shows the wave properties (like interference, diffraction etc.) are associated with it and when it is in the state of rest then it shows particle properties. Thus the matter has dual nature.
- Using the concept of the electron matter wave, de Broglie provided a rationale for the quantization of the electron's angular momentum in the hydrogen atom, which was postulated in Bohr's quantum theory.
- Compton's formula established that an electromagnetic wave can behave like a particle of light when interacting with matter. In 1924, Louis de Broglie proposed a new speculative hypothesis that electrons and other particles of matter can behave like waves. Today, this idea is known as de Broglie's hypothesis of matter waves.
- The De Broglie hypothesis proposes that all matter exhibits wave-like properties and relates the observed wavelength of matter to its momentum. After Albert Einstein's photon theory became accepted, the question became whether this was true only for light or whether material objects also exhibited wave-like behavior.
- De Broglie, in his 1924 PhD thesis, proposed that just as light has both wave-like and particle-like properties, electrons also have wave-like properties.

Davisson and Germer Experiment

- The Davisson-Germer experiment demonstrated the wave nature of the electron, confirming the earlier hypothesis of deBroglie.
- Putting wave-particle duality on a firm experimental footing, it represented a major step forward in the development of quantum mechanics.
- First conducted in 1927, the Davisson Germer Experiment was carried out by two scientists, C.J. Davisson and L.H. Germer, to explain the dual nature of electrons, i.e. Particle-wave Nature of the Electron.
- De Broglie proposed it in 1924, the dual nature of matter and the affirmation of Quantum Mechanics.
- Davisson and Germer Experiment, for the first time, proved the wave nature of electrons and verified the de Broglie equation.
- The basic thought behind the Davisson and Germer experiment was that the waves reflected from two different atomic layers of a Ni crystal will have a fixed phase difference.
- After reflection, these waves will interfere either constructively or destructively. Hence producing a diffraction pattern.
- In 1927 at Bell Labs, Clinton Davisson and Lester Germer fired slow moving electrons at a crystalline nickel target.

- The angular dependence of the reflected electron intensity was measured and was determined to have the same diffraction pattern as those predicted by Bragg for X-rays.

Phase velocity and group velocity

- Phase and group velocity are two important and related concepts in wave mechanics. They arise in quantum mechanics in the time development of the state function for the continuous case, i.e. wave packets.
- Phase velocity is the velocity at which the phase of a wave travels.
- Group velocity is equal to the energy at which the wave travels.
- In a dispersive medium, the phase velocity depends on the frequency of the wave. In a non-dispersive medium, the phase velocity of a wave is independent of the frequency of the given wave.
- The phase velocity is given by the equation $v_p = \omega/k$, where ω is the angular frequency and k is the wave number.
- The function $\omega(k)$, which gives ω as a function of k , is known as the dispersion relation. If ω is directly proportional to k , then the group velocity is exactly equal to the phase velocity. A wave of any shape will travel undistorted at this velocity.

Time-dependent and time-independent Schrodinger wave equations

The Schrodinger wave equation is a mathematical equation that describes how a quantum system changes over time. It is a fundamental equation in quantum mechanics and is used to predict the probability of finding a particle in a particular location at a given time.

There are two forms of the Schrodinger wave equation: the time-dependent form and the time-independent form.

The time-dependent form of the Schrodinger wave equation describes a system that evolves over time and is thus dependent on the physical state of that system. The time-dependent variant of Schrodinger's wave equation has the following expression: $i\hbar \frac{d}{dt} \psi(t) = \hat{H} \psi(t)$.

On the other hand, the time-independent Schrodinger equation (TISEq) can be derived from the equation that describes the motion of a wave in classical mechanics: (18.1.1) $\psi(x, t) = \exp [i(kx - \omega t)]$, where x is the position, t is time, $k = 2\pi/\lambda$ is the wave vector, and $\omega = 2\pi\nu$ is the angular frequency of the wave.

The time-dependent Schrödinger equation predicts that wave functions can form standing waves, called stationary states. These states are particularly important as their individual study later simplifies the task of solving the time-dependent Schrödinger equation for any state.

In the case where the Hamiltonian $H = \hat{p}^2/2m + V(x, t)$ is such that the potential $V(x, t)$ is time independent (we can then write $V(x)$), we can use

separation of variables to look for solutions.

Physical interpretation of wave function

- The wave function, at a particular time, contains all the information that anybody at that time can have about the particle.
- The wave function itself has no physical interpretation. It is not measurable.
- However, the square of the absolute value of the wave function has a physical interpretation.
- There is no physical meaning of wave function as it is not a quantity which can be observed. Instead, it is complex.
- For light, the wave-function has the physical meaning of propagating two orthogonal fields: an electric field and a magnetic field. This wave function is a real number.
- A clue to the physical meaning of the wavefunction $\Psi(x, t)$ is provided by the two-slit interference of monochromatic light that behaves as electromagnetic waves. The wavefunction of a light wave is given by $E(x, t)$, and its energy density is given by $|E|^2$, where E is the electric field strength.

Particle in a One-Dimensional Box

A particle in a one-dimensional box is a fundamental problem in quantum mechanics that demonstrates the discrete nature of energy levels in a confined system. It is a simple model that describes the behavior of a particle that is free to move in a one-dimensional space, but is confined by infinite potential barriers at the boundaries.

Some key points to remember about the particle in a one-dimensional box are:

1. The particle is free to move within the box, but cannot escape due to the infinite potential barriers at the boundaries.
2. The energy levels of the particle are discrete and quantized, meaning that the particle can only have certain allowed energy values.
3. The wave function of the particle must be zero at the boundaries of the box, due to the infinite potential barriers.
4. The allowed energy levels of the particle can be calculated using the Schrödinger equation, and are given by the formula: $E_n = (n^2 * h^2) / (8mL^2)$, where n is a positive integer, h is Planck's constant, m is the mass of the particle, and L is the length of the box.
5. The wave function of the particle can also be calculated, and is given by the formula: $\psi_n(x) = \sqrt{2/L} * \sin(n * \pi * x / L)$, where n is a positive integer, L is the length of the box, and x is the position of the particle within the box.

Unit 2 - Electromagnetic Field Theory

1. Electromagnetic field theory is the study of the interaction between electric charges and magnetic fields.
2. The fundamental principles of electromagnetic field theory are based on the laws of electromagnetism, which were first formulated by James Clerk Maxwell in the 19th century.
3. These laws describe the behavior of electric and magnetic fields, and their interactions with charged particles.
4. The four Maxwell's equations are the foundation of electromagnetic field theory. They are:
 - Gauss's law for electricity: The electric flux through any closed surface is proportional to the charge enclosed within the surface.
 - Gauss's law for magnetism: The magnetic flux through any closed surface is zero.
 - Faraday's law of induction: The electromotive force induced in a closed circuit is equal to the rate of change of the magnetic flux through the circuit.
 - Ampere's law with Maxwell's correction: The magnetic field around a closed loop is proportional to the current flowing through the loop, plus the displacement current.
5. Electromagnetic waves, such as light, radio waves, and microwaves, are a consequence of the interaction between electric and magnetic fields.
6. Electromagnetic field theory has numerous applications, including the design of electrical and electronic devices, communication systems, and medical imaging technologies.

Basic concept of Stoke's theorem and Divergence theorem for the notes of the Unit 2 - Electromagnetic Field Theory in the subject of ENGINEERING PHYSICS

Stoke's Theorem: - Stoke's theorem is a statement about the integration of differential forms on manifolds. - It relates the integral of a differential form over the boundary of some orientable manifold to the integral of its exterior derivative over the whole manifold. - Mathematically, it is expressed as: $\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S}$, where S is any surface bounded by curve C , $\mathbf{F}$ is a vector field, and $d\mathbf{S}$ and $d\mathbf{r}$ are the surface and line elements, respectively.

Divergence Theorem: - The divergence theorem, also known as Gauss's theorem, is a result that relates the flow of a vector field through a closed surface to the behavior of the vector field inside the surface. - It states that the outward flux of a vector field through a closed surface is equal to the volume integral of the divergence of the field within the surface. - Mathematically, it is expressed as:

$\iiint_V \text{div } \mathbf{F} \, dV = \iint_S \mathbf{F} \cdot d\mathbf{S}$, where V is any volume bounded by surface S , $\mathbf{F}$ is a vector field, and dV and $d\mathbf{S}$ are the volume and surface elements, respectively.

These theorems are fundamental concepts in vector calculus and are widely used

in the study of electromagnetism and fluid dynamics. They provide a way to relate the behavior of a vector field on the boundary of a region to its behavior within the region. In the context of electromagnetic field theory, they are used to derive important results such as Maxwell's equations.

Basic laws of electricity and magnetism for the notes of the Unit 2 - Electromagnetic Field Theory in the subject of ENGINEERING PHYSICS

1. **Coulomb's Law:** This law states that the force between two point charges is directly proportional to the product of the magnitudes of the charges and inversely proportional to the square of the distance between them. The force acts along the line joining the two charges.
2. **Ohm's Law:** This law states that the current flowing through a conductor is directly proportional to the potential difference applied across its ends, provided the physical conditions, such as temperature, remain constant.
3. **Ampere's Law:** This law states that the magnetic field produced by an electric current is proportional to the current and encircles the current-carrying conductor.
4. **Faraday's Law of Electromagnetic Induction:** This law states that a change in the magnetic field within a closed loop of wire induces an electromotive force (EMF) in the wire. This induced EMF is proportional to the rate of change of the magnetic flux through the loop.
5. **Lenz's Law:** This law states that the direction of the induced EMF and hence the induced current in a closed circuit is such as to oppose the change in magnetic flux that produces it.
6. **Biot-Savart Law:** This law states that the magnetic field produced by a current-carrying wire is proportional to the current and inversely proportional to the distance from the wire.
7. **Gauss's Law for Magnetism:** This law states that the net magnetic flux through any closed surface is zero. This implies that magnetic monopoles do not exist.

Continuity equation for current density for the notes of the Unit 2 - Electromagnetic Field Theory in the subject of ENGINEERING PHYSICS

- The continuity equation for current density is a mathematical expression that describes the conservation of electric charge.
- It states that the rate of change of electric charge in a volume is equal to the net current flowing into the volume.
- Mathematically, the continuity equation is expressed as: $\nabla \cdot \mathbf{J} + \frac{\partial \rho}{\partial t} = 0$, where $\mathbf{J}$ is the current density, ρ is the charge density, and t is time.

- The equation can be derived from the principle of conservation of charge, which states that the total electric charge in an isolated system is conserved.
- The continuity equation is an important concept in electromagnetism and is used to analyze the behavior of electric currents in various situations.
- It is also used in the study of electromagnetic waves and their interaction with matter.

Displacement current

- Displacement current is a phenomenon in electromagnetism that is analogous to an ordinary electric current. It is posited to explain magnetic fields that are produced by changing electric fields.
- Displacement current density is the quantity $\frac{dD}{dt}$ appearing in Maxwell's equations that is defined in terms of the rate of change of D , the electric displacement field.
- Displacement current density has the same units as electric current density, and it is a source of the magnetic field just as actual current is.
- A traveling and varying magnetic field is related to a synchronous change in the electric field that results in displacement current.
- The displacement current plays a major role during the propagation of electromagnetic radiations like light waves and radio waves through space.

Maxwell's Equations in Integral and Differential Form

Maxwell's equations are a set of four equations that describe the behavior of the electric and magnetic fields. These equations are named after James Clerk Maxwell, who first published them in 1861. The equations can be written in both integral and differential form.

1. **Gauss's Law for Electricity:** This law states that the electric flux through any closed surface is equal to the charge enclosed within the surface divided by the permittivity of free space. In integral form, it is written as:

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

In differential form, it is written as:

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

2. **Gauss's Law for Magnetism:** This law states that the magnetic flux through any closed surface is zero. In integral form, it is written as:

$$\oint \vec{B} \cdot d\vec{A} = 0$$

In differential form, it is written as:

$$\nabla \cdot \vec{B} = 0$$

3. **Faraday's Law of Induction:** This law states that the electromotive force (EMF) induced in a closed loop is equal to the negative rate of change of the magnetic flux through the loop. In integral form, it is written as:

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

In differential form, it is written as:

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

4. **Ampere's Law with Maxwell's Correction:** This law states that the magnetic field around a closed loop is equal to the sum of the current flowing through the loop and the displacement current, which is the rate of change of the electric flux through the loop. In integral form, it is written as:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(I + \epsilon_0 \frac{d\Phi_E}{dt} \right)$$

In differential form, it is written as:

$$\nabla \times \vec{B} = \mu_0 \left(\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right)$$

These equations are fundamental to the study of electromagnetic field theory and have numerous applications in engineering and physics. They provide a complete description of the behavior of electric and magnetic fields and their interactions with charges and currents. Understanding these equations is essential for the study of electromagnetic waves, optics, and electrical circuits.

Maxwell equations in vacuum and in conducting medium

Maxwell's equations are a set of four partial differential equations that describe the behavior of the electric and magnetic fields. These equations are used to derive the speed of electromagnetic waves in vacuum and non-conducting mediums.

In a vacuum, the current density, J , is equal to 0. Differentiating Maxwell's third equation on both sides gives us the speed of light (i.e. phase velocity) in a medium with permeability μ , and permittivity ϵ , and ∇^2 is the Laplace operator. In a vacuum, $v_{ph} = c_0 = 299792458 \text{ m/s}$, a fundamental physical constant.

Maxwell's equations are solved in homogenous mediums 1 and 2 separately. The solutions obtained by doing so are connected via the boundary conditions. In electromagnetic wave problems involving two mediums, boundary conditions for tangential electric fields and normal electric fields are applied to constrain the solutions.

At every point in space and at every time, the field vectors satisfy the Maxwell equations. where $\epsilon_0 = 8.8544 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$ is the dielectric permittivity and $\mu_0 = 1.2566 \times 10^{-6} \text{ m kg C}^{-2}$ is the magnetic permeability of vacuum.

Maxwell's equations may be combined to demonstrate how fluctuations in electromagnetic fields (waves) propagate at a constant speed in vacuum, $c(299792458 \text{ m/s})$. Known as electromagnetic radiation, these waves occur at various wavelengths to produce a spectrum of radiation from radio waves to gamma rays.

Maxwell equations in media can be modified by including to U and j & charges and currents inside atoms. In this case fields (so-called microscopic fields) are changing on microscopic scales comparable with atomic size and number of sources is too large.

Poynting Vector and Poynting Theorem

- The Poynting vector represents the directional energy flux or power flow of an electromagnetic field.
- The SI unit of the Poynting vector is the watt per square meter (W/m^2) or kg/s^3 in base SI units.
- The Poynting vector is used in conjunction with Poynting's theorem, the continuity equation expressing conservation of electromagnetic energy, to calculate the power flow in electric and magnetic fields.
- Poynting's theorem is a statement of conservation of energy for electromagnetic fields developed by British physicist John Henry Poynting.
- It states that in a given volume, the stored energy changes at a rate given by the work done on the charges within the volume, minus the rate at which energy leaves the volume.
- The cross product of the electric field vector, E , and the magnetic field vector, H , at any point is a measure of the rate of flow of electromagnetic energy per unit area at that point, that is $P = E \times H$. Here P represents the Poynting vector.
- The Poynting vector is named after its discoverer, J.H. Poynting.

Plane Electromagnetic Waves in Vacuum and Their Transverse Nature

Unit 2 - Electromagnetic Field Theory in the Subject of Engineering Physics

1. Electromagnetic waves are transverse waves that consist of oscillating electric and magnetic fields that propagate through space.
2. In a vacuum, these waves travel at the speed of light, which is approximately 299,792,458 meters per second.
3. The electric and magnetic fields of an electromagnetic wave are perpendicular to each other and to the direction of propagation of the wave.
4. This means that the electric and magnetic fields oscillate in planes that are perpendicular to the direction in which the wave is traveling.
5. The transverse nature of electromagnetic waves can be demonstrated using polarization filters, which only allow light waves with a specific orientation of the electric field to pass through.
6. When two polarization filters are placed at right angles to each other, no light can pass through, demonstrating that the electric field of the light wave is oscillating in a plane perpendicular to the direction of propagation.
7. Electromagnetic waves can have different frequencies and wavelengths, and the relationship between these two properties is given by the equation $c = f \lambda$, where c is the speed of light, f is the frequency of the wave, and λ is its wavelength.
8. The electromagnetic spectrum includes a wide range of frequencies and wavelengths, from radio waves with long wavelengths and low frequencies to gamma rays with short wavelengths and high frequencies.
9. Electromagnetic waves can be generated by accelerating electric charges, and they can interact with matter in various ways, including absorption, reflection, and refraction.
10. The study of electromagnetic waves and their interactions with matter is an important part of the field of electromagnetism and has many practical applications in areas such as communication, medicine, and energy production.

Relation between electric and magnetic fields of an electromagnetic wave

1. Electromagnetic waves are transverse waves that consist of oscillating electric and magnetic fields that are perpendicular to each other and to the direction of propagation of the wave.
2. The electric and magnetic fields of an electromagnetic wave are in phase and their magnitudes are related by the equation $E = cB$, where E is the electric field strength, B is the magnetic field strength, and c is the speed of light.
3. The electric and magnetic fields of an electromagnetic wave are mutually dependent and self-sustaining. A changing electric field generates a mag-

- netic field, and a changing magnetic field generates an electric field.
4. The energy of an electromagnetic wave is equally distributed between the electric and magnetic fields.
 5. The Poynting vector, which represents the direction and magnitude of the energy flow of an electromagnetic wave, is given by the cross product of the electric and magnetic fields: $\mathbf{S} = \mathbf{E} \times \mathbf{H}$, where $\mathbf{H}$ is the magnetic field intensity.
 6. The electric and magnetic fields of an electromagnetic wave obey the wave equation, which describes how the fields propagate through space and time.
 7. The electric and magnetic fields of an electromagnetic wave are described by Maxwell's equations, which relate the electric and magnetic fields to their sources (charges and currents) and to each other.

Plane Electromagnetic Waves in Conducting Medium

- Electromagnetic waves are transverse waves that consist of oscillating electric and magnetic fields.
- In a conducting medium, the electric field generates a current that produces a magnetic field, and the magnetic field generates an electric field.
- The electric and magnetic fields are perpendicular to each other and to the direction of wave propagation.
- The speed of the wave in a conducting medium is determined by the conductivity and permittivity of the medium.
- The wave is attenuated as it travels through the medium due to the resistance of the medium.
- The skin depth is the distance over which the amplitude of the wave decreases by a factor of $1/e$. It is determined by the conductivity and frequency of the wave.
- The wave impedance is the ratio of the electric field to the magnetic field and is determined by the conductivity and permittivity of the medium.
- The reflection and transmission of the wave at the interface between two media are determined by the wave impedance of the two media.

Skin Depth

Skin depth is an important concept in Electromagnetic Field Theory. It is defined as the distance over which the magnitude of the electric or magnetic field is reduced by a factor of $1/e$ (about 37%) of the value at the surface. This distance is referred to as the skin depth of the conductor.

- Skin depth is represented by the symbol δ .
- It depends on the frequency of the current and the electrical and magnetic properties of the conductor.
- An electromagnetic wave entering a conducting surface is damped and reduces in amplitude by a factor of $1/e$ in a distance δ .

- Theoretically, for eddy current, skin depth is given by the formula: $1/\sqrt{f}$, where f is the coil frequency, μ is the magnetic permeability (in H/mm) of the material, and σ is the electrical conductivity (in %IACS) of the material.

Unit 3 - Wave Optics

Wave optics, also known as physical optics, is the study of light as a wave. It is concerned with the behavior of light when it encounters obstacles or interacts with other waves. Some of the phenomena that can be explained using wave optics include interference, diffraction, and polarization.

1. **Interference:** Interference occurs when two or more waves meet and interact with each other. The resulting wave can have a larger or smaller amplitude than the individual waves, depending on how they combine. This can result in constructive or destructive interference.
2. **Diffraction:** Diffraction is the bending of light around obstacles or through small openings. It occurs because light behaves as a wave and can spread out as it passes through an opening or around an obstacle.
3. **Polarization:** Polarization is the orientation of the electric field of a light wave. Light can be polarized by passing it through a polarizing filter, which only allows light with a certain orientation of the electric field to pass through.

Coherent sources for the notes of the Unit 3 - Wave Optics in the subject of ENGINEERING PHYSICS

- Coherent sources are sources of light that emit waves of the same frequency and constant phase difference.
- The waves from coherent sources interfere constructively and destructively to produce an interference pattern.
- Coherent sources can be created by dividing a single light source into two or more beams using a beam splitter or by using two or more lasers that are phase-locked.
- Examples of coherent sources include lasers, masers, and some gas discharge lamps.
- Coherent sources are important in many applications, including holography, interferometry, and optical communications.

Interference in Uniform and Wedge Shaped Thin Films

Interference is a phenomenon in which two or more waves superimpose to form a resultant wave of greater, lower, or the same amplitude. Interference usually refers to the interaction of waves that are correlated or coherent with each other, either because they come from the same source or because they have the same or nearly the same frequency.

In the context of uniform and wedge-shaped thin films, interference occurs when light waves reflect off the top and bottom surfaces of a thin film and then combine. The interference can be either constructive or destructive, depending on the thickness of the film, the angle of incidence, and the wavelength of the light.

1. **Uniform Thin Films:** In a uniform thin film, the thickness of the film is constant. When light is incident on the film, some of it is reflected at the top surface, while the rest is transmitted into the film. The transmitted light is then partially reflected at the bottom surface of the film, and the two reflected waves combine to produce interference.
2. **Wedge-Shaped Thin Films:** In a wedge-shaped thin film, the thickness of the film varies. This means that the path difference between the two reflected waves also varies, leading to a variation in the interference pattern. This can result in the formation of fringes, which are bands of light and dark regions.

Necessity of extended sources for the notes of the Unit 3 - Wave Optics in the subject of ENGINEERING PHYSICS

1. Extended sources provide a more comprehensive understanding of the subject matter.
2. They offer a wider range of perspectives and approaches to the topic.
3. They can provide additional examples and explanations to clarify complex concepts.
4. They can offer alternative methods of problem-solving and analysis.
5. They can provide additional practice problems and exercises to reinforce understanding.
6. They can provide up-to-date information and developments in the field.
7. They can provide a broader context for the subject matter, connecting it to other areas of study.
8. They can provide additional resources for further study and exploration of the topic.

Overall, the use of extended sources in the study of Unit 3 - Wave Optics in the subject of ENGINEERING PHYSICS can greatly enhance the learning experience and deepen understanding of the subject matter. It is highly recommended to incorporate extended sources into the study of this unit.

Newton's Rings and its applications

Newton's Rings is a phenomenon in which an interference pattern is created by the reflection of light between two surfaces - a spherical surface and an adjacent flat surface. It is named after Sir Isaac Newton, who first studied the effect in the 17th century.

The phenomenon is observed when a plano-convex lens is placed on a flat glass

surface, and monochromatic light is shone onto the lens from above. The light reflects off the bottom surface of the lens and the top surface of the glass, creating an interference pattern of bright and dark rings.

The radius of the rings is given by the formula: $R_n = \sqrt{(R^2 - (n \lambda/2))}$, where R is the radius of curvature of the lens, n is the order of the ring, and λ is the wavelength of the light.

Applications of Newton's Rings include: - Determining the wavelength of light: By measuring the radii of the rings and using the above formula, the wavelength of the light can be calculated. - Testing the quality of optical surfaces: The presence of any defects or irregularities on the surfaces will cause distortions in the interference pattern, allowing for the detection of such defects. - Measuring the refractive index of a liquid: By placing a liquid between the lens and the glass surface, the interference pattern will change, allowing for the calculation of the refractive index of the liquid.

This phenomenon is an important concept in the study of wave optics and has numerous practical applications in the field of engineering physics. It is covered in Unit 3 - Wave Optics of the subject of ENGINEERING PHYSICS.

Introduction to Diffraction

Diffraction is the phenomenon of bending of light around the corners of an obstacle or aperture into the region of geometrical shadow of the obstacle. It is a characteristic property of waves, including light waves, sound waves, and water waves. Diffraction is observed when the size of the obstacle or aperture is comparable to the wavelength of the wave.

Some key points to remember about diffraction are: - Diffraction is a wave property and is observed for all types of waves. - The amount of diffraction depends on the size of the obstacle or aperture relative to the wavelength of the wave. - The smaller the obstacle or aperture, the more the wave spreads out. - The longer the wavelength of the wave, the more it spreads out.

In the context of wave optics, diffraction is an important phenomenon as it provides evidence for the wave nature of light. It is also used in various practical applications such as diffraction gratings and X-ray diffraction.

Fraunhofer Diffraction at Single Slit and Double Slit

Unit 3 - Wave Optics in the subject of ENGINEERING PHYSICS

1. **Fraunhofer Diffraction:** Fraunhofer diffraction is a type of diffraction that occurs when a wave passes through an aperture or around an obstacle, and the diffracted wavefronts are essentially parallel. This type of diffraction is also known as far-field diffraction.
2. **Single Slit Diffraction:** When a parallel beam of monochromatic light passes through a narrow slit, the light spreads out and forms a diffraction

pattern on a screen placed behind the slit. This phenomenon is known as single slit diffraction.

3. **Double Slit Diffraction:** Double slit diffraction is a phenomenon where light passing through two closely spaced slits produces an interference pattern on a screen placed behind the slits. This pattern consists of a series of bright and dark fringes.
4. **Difference between Single and Double Slit Diffraction:** The main difference between single and double slit diffraction is that in single slit diffraction, the diffraction pattern is formed due to the interference of light waves diffracted from a single slit, while in double slit diffraction, the diffraction pattern is formed due to the interference of light waves diffracted from two closely spaced slits.
5. **Applications of Single and Double Slit Diffraction:** Single and double slit diffraction have many applications in various fields such as optics, physics, and engineering. Some of the common applications include the determination of the wavelength of light, the study of the properties of light, and the measurement of the size of small objects.

Absent Spectra

- Absent spectra, also known as absorption spectra, are produced when light passes through a cool gas.
- The gas absorbs certain wavelengths of light, leaving dark lines in the spectrum.
- These dark lines correspond to the wavelengths of light that have been absorbed by the gas.
- The pattern of dark lines is unique to each element, allowing scientists to identify the elements present in a gas by analyzing its absorption spectrum.
- Absent spectra are used in a variety of fields, including astronomy, chemistry, and environmental science.
- In astronomy, for example, scientists can use the absorption spectra of stars to determine their chemical composition.
- In chemistry, absorption spectra can be used to identify the presence of certain elements in a sample.
- In environmental science, absorption spectra can be used to measure the concentration of pollutants in the air or water.
- Absent spectra are an important tool for scientists in many different fields, allowing them to analyze and understand the world around us.

Diffraction Grating

A diffraction grating is an optical component with a periodic structure that splits and diffracts light into several beams traveling in different directions. The directions of these beams depend on the spacing of the grating and the wavelength of

the light so that the grating acts as the dispersive element. Diffraction gratings are used in spectrometers, monochromators, and laser pulse compression.

- A diffraction grating can be manufactured by ruling a large number of closely spaced parallel lines on a glass or metal surface.
- The spacing between the lines, known as the grating constant, determines the angles at which the light is diffracted.
- The grating equation relates the angles of incidence and diffraction to the grating constant and the wavelength of the light.
- Diffraction gratings can be used to separate light of different wavelengths, producing a spectrum.
- They can also be used to measure the wavelength of light by observing the angles at which the light is diffracted.
- Diffraction gratings are commonly used in spectroscopy, where they are used to analyze the spectral composition of light.

Spectra with grating for the notes of the Unit 3 - Wave Optics in the subject of ENGINEERING PHYSICS

- A diffraction grating is an optical component with a periodic structure that splits and diffracts light into several beams traveling in different directions.
- The directions of these beams depend on the spacing of the grating and the wavelength of the light so that the grating acts as the dispersive element.
- The diffraction pattern formed with a grating is known as a grating spectrum.
- The spectrometer uses a prism or a grating to spread the light into a spectrum.
- This allows astronomers to detect many of the chemical elements by their characteristic spectral lines.
- These lines are named for the elements which cause them, such as the hydrogen alpha, beta, and gamma lines.
- Diffraction gratings are commonly used for spectroscopic dispersion and analysis of light.
- What makes them particularly useful is the fact that they form a sharper pattern than double slits do.
- That is, their bright fringes are narrower and brighter while their dark regions are darker.
- If the number of lines in the grating are very large then the maxima are sharp and bright.

Dispersive Power

Dispersive power is a property of a material that determines the degree of separation of light of different colors (wavelengths) when it passes through the material. It is defined as the ratio of the difference in the angles of deviation of two light rays of different wavelengths to the difference in the angles of minimum deviation of the same two rays.

Mathematically, the dispersive power () of a material is given by the formula:

$$= (v - r) / (y - 1)$$

where v , r , and y are the refractive indices of the material for violet, red, and yellow light, respectively.

Dispersive power is an important property of optical materials, as it determines the degree of chromatic aberration in optical instruments. Materials with high dispersive power can separate light into its constituent colors more effectively, but also produce more chromatic aberration.

In summary, dispersive power is a measure of the ability of a material to separate light into its constituent colors. It is determined by the refractive indices of the material for different wavelengths of light and is an important property to consider in the design of optical instruments.

Resolving Power

Resolving power is the ability of an optical instrument to distinguish between two closely spaced objects. It is a measure of the clarity of an image produced by the instrument. The resolving power of an optical instrument is determined by the wavelength of the light being used and the numerical aperture of the instrument.

Here are some key points to remember about resolving power:

1. The resolving power of an optical instrument is inversely proportional to the wavelength of the light being used. This means that shorter wavelengths of light, such as blue or violet, will produce images with higher resolving power than longer wavelengths, such as red or orange.
2. The numerical aperture of an optical instrument is a measure of its ability to gather light. The higher the numerical aperture, the greater the resolving power of the instrument.
3. The resolving power of an optical instrument can be improved by using light of shorter wavelengths or by increasing the numerical aperture of the instrument.
4. The Rayleigh criterion is a commonly used measure of the resolving power of an optical instrument. According to this criterion, two objects are considered to be resolved if the central maximum of the diffraction pattern of one object falls on the first minimum of the diffraction pattern of the other object.
5. The resolving power of the human eye is limited by the spacing of the photoreceptor cells in the retina. The average resolving power of the human eye is about 0.1 arc minutes.

Rayleigh's criterion of resolution

Rayleigh's criterion of resolution is a standard for the minimum separation between two light sources that can be resolved by an optical instrument. It is based on the diffraction pattern produced by the instrument's aperture.

1. According to Rayleigh's criterion, two point sources are considered to be just resolved when the central maximum of the diffraction pattern of one source coincides with the first minimum of the diffraction pattern of the other source.
2. The minimum angular separation between the two sources that can be resolved is given by the formula $\theta = 1.22 \lambda / D$, where λ is the wavelength of the light and D is the diameter of the aperture.
3. The Rayleigh criterion is commonly used to determine the resolving power of optical instruments such as telescopes and microscopes.
4. The criterion is named after the British scientist Lord Rayleigh, who first derived it in the 19th century.
5. It is important to note that the Rayleigh criterion is a theoretical limit and that in practice, the resolution of an optical instrument may be limited by other factors such as the quality of the optics and the sensitivity of the detector.

Resolving Power of Grating

- The resolving power of a grating is defined as its ability to separate two closely spaced wavelengths of light.
- It is given by the formula: $R = nN$, where R is the resolving power, n is the order of the spectrum, and N is the total number of lines on the grating.
- The resolving power of a grating increases with the order of the spectrum and the number of lines on the grating.
- The resolving power of a grating is also affected by the width of the slits and the distance between them.
- A grating with narrow slits and a large number of lines will have a higher resolving power than a grating with wider slits and fewer lines.
- The resolving power of a grating can be improved by using a collimator to produce parallel light rays and by using a telescope to magnify the image of the spectrum.
- The resolving power of a grating is important in spectroscopy, where it is used to separate and analyze the different wavelengths of light emitted or absorbed by a substance.

Unit 4 - Fiber Optics & Laser

Fiber optics and lasers are two important technologies that have revolutionized the field of telecommunications and medicine. Here are some key points to know about these technologies:

1. **Fiber Optics** is a technology that uses thin, flexible fibers of glass or plastic to transmit data as light signals over long distances. These fibers are bundled together to form a cable, which can carry a large amount of data at high speeds.
2. **Lasers** are devices that produce intense beams of light. They work by stimulating the emission of light from atoms or molecules, resulting in a highly focused and coherent beam of light.
3. Fiber optic cables are used in telecommunications to transmit data over long distances. They have several advantages over traditional copper cables, including higher bandwidth, longer transmission distances, and immunity to electromagnetic interference.
4. Lasers have many applications in medicine, including surgery, diagnostics, and therapy. They can be used to cut, vaporize, or coagulate tissue, and are often used in procedures such as LASIK eye surgery and cancer treatment.
5. Both fiber optics and lasers rely on the properties of light and its interaction with matter. Understanding the principles of optics and the behavior of light is essential to understanding these technologies.
6. Fiber optics and lasers have revolutionized many fields, and their applications continue to expand as new technologies and techniques are developed. They are essential tools in modern telecommunications, medicine, and many other fields.

Fibre Optics: Principle and construction of optical fiber

Fiber optics is the science of transmitting data, voice, and images by the passage of light through thin, transparent fibers. Optical fibers typically work on the principle of total internal reflection of light. The term total internal reflection indicates that no part of the signal gets refracted or transmitted to the second medium, but instead, all of the signals is retained inside the fiber.

Modern optical fibers are made of double layers of glass. The inner fiber core (8 μm) is encircled by cladding (125 μm), a concentric core of lower index glass. A protective layer surrounds the cladding. The core-cladding interface is where total internal reflection occurs. The transmission of light along optical fibers is based on the total internal reflection theory.

Optical fibers have many advantages over traditional copper cables. They transmit information far more quickly and have a higher bandwidth. In telecommunications, fiber optic technology has virtually replaced copper wire in long-distance telephone lines, and it is used to link computers within local area networks.

Acceptance angle

- The acceptance angle of an optical fiber is defined based on a purely geometrical consideration (ray optics): it is the maximum angle of a ray (against the fiber axis) hitting the fiber core which allows the incident light to be guided by the core.
- The maximum incidence angle of a light ray which can be used for injecting light into a fiber core or waveguide is known as the Acceptance Angle.
- Acceptance angle is just the conical half angle of the acceptance cone.
- Acceptance angle is the maximum angle with the axis of the optical fiber at which light can enter the fiber, in order to be propagated through it.

Numerical Aperture

Numerical aperture (NA) is a measure of the angular acceptance of an optical system, such as an imaging system or a fiber optic. It is defined based on geometrical considerations and is a theoretical parameter calculated from the optical design .

In fiber optics, the numerical aperture describes the range of angles within which light that is incident on the fiber will be transmitted along it . It is considered as the light-gathering capacity of an optical fiber. The numerical aperture is defined as the sine of half of the angle of the fiber's light acceptance cone, i.e., $NA = \sin(a)$, where a is called the acceptance cone angle .

In laser physics, the numerical aperture is defined slightly differently. Laser beams spread out as they propagate, but slowly. Far away from the narrowest part of the beam, the spread is roughly linear with distance, forming a cone of light in the “far field” .

Acceptance Cone

- The acceptance cone is the cone within which optical power may be coupled into the bound modes of an optical fiber .
- The cross section of an optical fiber is circular; the light waves accepted by the core are expressed as a cone .
- The larger the acceptance cone, the larger the numerical aperture (NA); this means that the fiber is able to accept and propagate more light .
- The acceptance cone is derived by rotating the acceptance angle about the fiber axis .
- In order to get high light transmission, light should propagate through the fiber by a series of total internal reflections. This transmission is achieved if the angles of the light entering the fiber are within the acceptance cone .

Step Index and Graded Index Fibers

Step index fiber and graded index fiber are two types of optical fibers used in fiber optics communication. Here are some key points to note about these two types of fibers:

1. **Step index fiber** is a fiber in which the core is of a uniform refractive index and there is a sharp decrease in the index of refraction at the cladding.
2. **Graded index fiber** is a type of fiber where the refractive index of the core is maximum at the center core and then it decreases towards core-cladding interface.
3. One of the major differences between a step index fiber and a graded index fiber is that a step index fiber has a core with constant refractive index and makes a step change from the core – cladding interface while a graded index fiber does not have a core with constant refractive index, and it makes a gradual change from the core – cladding interface.
4. Step index fiber is often used for short distance communication. Step index fibers are less expensive to manufacture than graded index fibers. They have a higher numerical aperture, this allows them to collect light from a wider range of angles. Step index fibers have higher dispersion.
5. Step index fiber is of two types viz; mono mode fiber and multi mode fiber. Graded index fiber is of only of one type that is multi mode fiber.
6. In step index fiber, different modes are launched at different acceptance angles causing higher modes to travel longer paths. This causes modal dispersion. In graded index fiber, the refractive index of the core gradually lowers towards the cladding. Light travels faster in lower refractive index.

Fiber Optic Communication Principle

Fiber optic communication is a method of transmitting information from one place to another by sending pulses of infrared light through an optical fiber. The light is a form of carrier wave that is modulated to carry information.

The fiber optic works on the principle of total internal reflection. When a ray of light is incident on the core of the optical fiber at a small angle, it suffers refraction and strikes the core cladding interface. As the diameter of the fiber is very small, the angle of incidence is greater than the critical angle.

Optical fiber is used by telecommunications companies to transmit telephone signals, Internet communication, and cable television signals. It is also used in other industries, including medical, defense, government, industrial, and commercial.

First, an electrical signal is converted into an optical signal, and then an optical signal is transmitted through the optical fiber, which is a type of wired communication.

The actual number of modes that can be propagated in a given optical fiber is determined by the wavelength of light and the diameter and index of refraction

of the core of the fiber. Signals lose strength as they are propagated through the fiber; this is known as beam attenuation.

Attenuation

Attenuation is the reduction in the strength of a signal as it travels through a medium. In the context of fiber optics, attenuation refers to the loss of light power as it travels through an optical fiber. This loss is measured in decibels (dB) and is typically expressed as a loss per unit length, such as dB/km.

There are several factors that contribute to attenuation in optical fibers, including:

1. **Absorption:** This occurs when the light energy is absorbed by the atoms in the glass, converting it into heat. This is primarily caused by impurities in the glass, such as water or transition metals.
2. **Scattering:** This occurs when light is scattered in different directions due to microscopic variations in the glass. This can be caused by fluctuations in the density or composition of the glass.
3. **Bending:** When an optical fiber is bent, some of the light can escape from the core and be lost. This is known as bend loss or macrobend loss.
4. **Splices and Connectors:** When two fibers are joined together, either through a splice or a connector, some light can be lost at the point of connection.

Attenuation is an important factor to consider when designing a fiber optic communication system, as it determines the maximum distance that a signal can travel before it becomes too weak to be detected. To minimize attenuation, optical fibers are carefully designed and manufactured to minimize impurities and variations in the glass, and splices and connectors are carefully aligned and polished to minimize loss.

Dispersion

Dispersion is a phenomenon that occurs when light travels through a medium and its different wavelengths are separated due to their different speeds. This can be observed in a prism, where white light is split into its constituent colors. In the context of fiber optics, dispersion can cause distortion of the signal being transmitted.

There are two main types of dispersion in fiber optics: chromatic dispersion and modal dispersion.

1. **Chromatic Dispersion:** This type of dispersion occurs due to the different speeds of different wavelengths of light in the fiber. It can be further divided into two types: material dispersion and waveguide dispersion. Material dispersion is caused by the variation of the refractive index of the

fiber material with wavelength. Waveguide dispersion is caused by the dependence of the propagation constant on the wavelength of light.

2. **Modal Dispersion:** This type of dispersion occurs when the light travels in different modes in the fiber. Each mode has a different propagation speed, causing the signal to spread out over time. Modal dispersion is more significant in multimode fibers than in single-mode fibers.

Dispersion can be minimized by using dispersion-shifted fibers, which are designed to have a zero-dispersion wavelength that matches the wavelength of the transmitted light. Dispersion compensators can also be used to correct for dispersion in the fiber.

In summary, dispersion is an important factor to consider in fiber optic communication systems, as it can cause distortion of the transmitted signal. Various techniques can be used to minimize its effects and ensure the integrity of the transmitted data.

Application of fiber

Fiber optics is the science of transmitting data, voice, and images by the passage of light through thin, transparent fibers. It has a wide range of applications due to its inherent advantages over copper conductors. Some of the major application areas of optical fibers are:

1. **Communications** - Voice, data, and video transmission are the most common uses of fiber optics. It has virtually replaced copper wire in long-distance telephone lines and is used to link computers within local area networks .
2. **Medical Industry** - Optical fibers are used in the medical industry for a variety of applications, including endoscopy, light therapy, and surgical procedures .
3. **Defense** - Fiber optics are used for data transmission in high-level data security fields of military and aerospace applications. These are used in wirings in aircraft, hydrophones for SONARs and Seismics applications .
4. **Industries** - Optical fibers are used in industries for a variety of applications, including industrial control systems and avionic systems .
5. **Broadcasting** - Fiber optics are used in broadcasting for the transmission of high-quality audio and video signals .
6. **Lighting and Decorations** - Optical fibers are used for lighting and decorations, including architectural lighting, landscape lighting, and decorative lighting .
7. **Mechanical Inspections** - Optical fibers are used for mechanical inspections, including the inspection of hard-to-reach areas and the inspection of mechanical components .

Laser: Absorption of radiation

1. Absorption of radiation is the process by which an atom, molecule or ion absorbs electromagnetic radiation and undergoes a transition from a lower energy state to a higher energy state.
2. The energy of the absorbed photon is equal to the difference in energy between the two states.
3. The absorption of radiation is a fundamental process in the operation of a laser.
4. In a laser, the active medium is pumped with energy, usually in the form of light or electrical current, to create a population inversion.
5. This means that there are more atoms or molecules in an excited state than in the ground state.
6. When a photon of the right energy passes through the active medium, it can stimulate an excited atom or molecule to drop to a lower energy state and emit a photon of the same energy.
7. This process is called stimulated emission and is the basis for laser amplification.
8. The emitted photon is in phase with the stimulating photon, resulting in coherent light.
9. The absorption of radiation is also important in the attenuation of laser light as it passes through a medium.
10. The absorption coefficient of a material determines how much laser light is absorbed as it passes through the material.
11. The absorption of laser light can also be used for spectroscopic analysis, as different materials absorb light at different wavelengths.
12. This can be used to identify the composition of a material or to measure the concentration of a particular substance.

Spontaneous and stimulated emission of radiation

Spontaneous and stimulated emission are two processes that can occur when an atom or molecule is in an excited state.

- **Spontaneous emission** takes place without interaction with other photons, and the direction and phase are random. When the excitation involves the absorption of radiation, the spontaneous emission is fluorescence. The atom in the excited state jumps back to the ground state by emitting a photon of energy without the aid of external agency after 10^{-8} s. Such emission of radiation is known as spontaneous emission. This process is random and the emitted radiation is incoherent.
- **Stimulated emission** takes place when the excited electron interacts with another photon. The notable characteristic of stimulated emission compared to everyday light sources (which depend on spontaneous emission) is that the emitted photons have the same frequency, phase, polarization, and direction of propagation as the incident photons. The photons

involved are thus mutually coherent.

These two processes are important in the study of lasers and fiber optics, as they play a crucial role in the generation of laser light and the transmission of information through optical fibers.

Population Inversion

- Population inversion is a technique mainly used for light amplification .
- It is required for laser operation .
- Consider a group of electrons with two energy levels E_1 and E_2 .
- A population inversion ($N_2 > N_1$) has thus been achieved between level 1 and 2, and optical amplification at the frequency ν can be obtained .
- At least half the population of atoms must be excited from the ground state to obtain a population inversion, the laser medium must be very strongly pumped .
- Population inversion, and thus optical gain in the sense of Eq. 14, may be obtained in a direct bandgap semiconductor exhibiting radiative recombination of electrons and holes provided there is, simultaneously in space and time, a large density of electrons in the conduction band and holes in the valence band such as that provided by the junction region of a diode under heavy forward bias where large numbers of electrons and holes exist .
- Population inversion and laser action have been discovered in a large number of optical transitions in gases and gas mixtures .
- A gas as the laser active medium has several advantages: scalability to large volume and effective cooling by diffusion or by convective flow, allowing potentially high power output .
- Normally, a system of atoms is in temperature equilibrium and there are always more atoms in low energy states than in higher ones .

Einstein's Coefficients

- Einstein coefficients are mathematical quantities that measure the probability of absorption or emission of light by an atom or molecule .
- The quantities A_{21} , B_{12} , and B_{21} are called Einstein's coefficients. These coefficients are determined by the atomic system .
- The Einstein A coefficients are related to the rate of spontaneous emission of light, while the Einstein B coefficients are related to the absorption and stimulated emission of light .
- The ratio of Einstein's coefficient of spontaneous emission to the Einstein's coefficient of stimulated absorption is proportional to the cube of frequency ν .
- Einstein's theory of absorption and emission of light by an atom is based on Planck's theory of radiation. Under thermal equilibrium, the population of energy levels obeys the Boltzmann's distribution law .

- The Einstein coefficients are fixed probabilities per time associated with each atom and do not depend on the state of the gas of which the atoms are a part. Therefore, any relationship that can be derived between the coefficients at, say, thermodynamic equilibrium will be valid universally .

Principles of laser action for the notes of the Unit 4 - Fiber Optics & Laser in the subject of ENGINEERING PHYSICS

1. **Population Inversion:** For laser action to occur, the number of atoms in the higher energy state must be greater than the number of atoms in the lower energy state. This condition is known as population inversion.
2. **Optical Feedback:** Laser action requires a feedback mechanism to sustain the stimulated emission of radiation. This is achieved by placing the lasing medium between two mirrors, forming an optical cavity. The mirrors reflect the emitted radiation back into the lasing medium, stimulating further emission.
3. **Pumping:** To achieve population inversion, the atoms in the lasing medium must be excited to the higher energy state. This is achieved by pumping energy into the lasing medium, typically using an electrical discharge or an intense light source.
4. **Gain:** The gain of a laser is the amplification of the emitted radiation as it passes through the lasing medium. The gain must be greater than the losses in the optical cavity for laser action to occur.
5. **Threshold:** The threshold of a laser is the minimum amount of pumping energy required to achieve population inversion and initiate laser action.
6. **Coherence:** The emitted radiation from a laser is highly coherent, meaning that the waves are in phase and have a well-defined direction. This is a result of the stimulated emission process, where the emitted radiation is in phase with the stimulating radiation.
7. **Monochromaticity:** The emitted radiation from a laser is highly monochromatic, meaning that it has a very narrow range of wavelengths. This is a result of the narrow energy levels in the lasing medium and the selective feedback provided by the optical cavity.
8. **Directionality:** The emitted radiation from a laser is highly directional, meaning that it is emitted in a well-defined direction. This is a result of the optical cavity, which only allows radiation traveling in a specific direction to be amplified and emitted.
9. **Intensity:** The emitted radiation from a laser is highly intense, meaning that it has a high power per unit area. This is a result of the amplification of the emitted radiation as it passes through the lasing medium multiple times.

Solid state Laser (Ruby laser) and Gas Laser (He-Ne laser)

Solid state Laser (Ruby laser)

- A ruby laser is a three-level solid-state laser.
- The gain medium is a solid at room temperature.
- Optical pumping technique is used to supply energy to the laser medium.
- Optical pumping is a technique in which light is used as an energy source to raise electrons from a lower energy level to a higher energy level.
- In solid-state laser materials, the atoms responsible for generating laser light are first excited to higher energy states through the absorption of photons.
- The way in which these atoms relax from their excited states determines the quality and quantity of laser light produced.

Gas Laser (He-Ne laser)

- A gas laser is used to create a continuous laser beam.
- High coherence, high directionality, and high monochromaticity beams can be achieved using gas lasers.
- The output power of a gas laser is usually measured in milliwatts.
- The first He-Ne gas laser was developed by Ali Javan and coworkers.
- The He-Ne laser has a wavelength of 632.8 nm and a linewidth of only 10kHz.

Laser applications for the notes of the Unit 4 - Fiber Optics & Laser in the subject of ENGINEERING PHYSICS

1. **Laser cutting and welding:** Fiber lasers have proven to be efficient for laser cutting and welding .
2. **Laser marking and engraving:** Fiber lasers are also efficient for laser marking and engraving .
3. **Laser cleaning:** Fiber lasers have proven to be efficient for laser cleaning .
4. **Sensing:** Detection and sensing technologies use optical fibers and their properties to collect data. Fiber optic sensing applications include LiDAR and spectroscopy .
5. **Medical applications:** Fiber lasers in medicine are one of the most meaningful applications of fiber laser technology. The great versatility of fiber lasers make them very effective for both surgery and diagnostic imaging, providing a high level of accuracy, precision and safety .
6. **Flexible delivery:** An advantage of fiber lasers over other types of lasers is that the laser light is both generated and delivered by an inherently flexible medium, which allows easier delivery to the focusing location and target. This can be important for laser cutting, welding, and folding of metals and polymers .

Unit 5 - Superconductors and Nano-Materials:

Superconductors are materials that exhibit zero electrical resistance and the expulsion of magnetic fields below a certain temperature, known as the critical temperature. This phenomenon is known as superconductivity.

- Superconductivity was first discovered in 1911 by Heike Kamerlingh Onnes, who observed that the resistance of mercury dropped to zero at a temperature of 4.2 K.
- Superconductors have a wide range of applications, including in MRI machines, particle accelerators, and power transmission lines.
- The critical temperature of a superconductor varies depending on the material, with some materials exhibiting superconductivity at temperatures as high as 138 K.
- The mechanism behind superconductivity is not yet fully understood, but it is believed to involve the formation of Cooper pairs, in which two electrons form a bound state and move through the material without resistance.

Nano-materials are materials with at least one dimension on the nanoscale, typically between 1 and 100 nanometers.

- Nano-materials have unique properties due to their small size and high surface area to volume ratio.
- They have a wide range of applications, including in electronics, medicine, and energy production.
- Some common nano-materials include carbon nanotubes, graphene, and quantum dots.
- The properties of nano-materials can be tuned by changing their size, shape, and composition.
- The study of nano-materials is a rapidly growing field, with new discoveries and applications being made regularly.

Superconductors: Temperature dependence of resistivity in superconducting materials

- Superconductors are special materials in which after a certain critical temperature the resistance drops to zero value and the conductivity thus reaches the maximum .
- The critical temperature is known as T_C .
- These temperatures are low, with most industrial superconductors below 12 K and higher temperature ceramics just above 130K .
- The superconductivity of material is influenced by external temperature, current, and external magnetic field .
- The resistivity of materials depend on the temperature. $\rho = \rho_0 [1 + \alpha(T - T_0)]$ is the equation that shows the relation between the temperature and the resistivity of a material. In the equation ρ_0 is the resistivity at a standard temperature, ρ is the resistivity at t °C, T_0 is the reference

temperature and α is the temperature coefficient of resistance .

Meissner effect

- The Meissner effect is the expulsion of a magnetic field from the interior of a material that is in the process of becoming a superconductor.
- This occurs when the material is cooled below a certain temperature, called the transition temperature, usually close to absolute zero.
- The Meissner effect is a property of all superconductors and was discovered by the German physicists W. Meissner and R. Ochsenfeld in 1933.
- As a superconductor in a magnetic field is cooled to the temperature at which it abruptly loses electrical resistance, all or part of the magnetic field within the material is expelled.
- One of the most well-known demonstrations of the Meissner effect is its ability to make a magnet levitate above a superconductor.

Temperature dependence of critical field

The critical field of a superconductor is the magnetic field strength at which the superconductor transitions from the superconducting state to the normal state. The critical field is dependent on the temperature of the superconductor.

- At absolute zero, the critical field is at its maximum value, known as the thermodynamic critical field.
- As the temperature of the superconductor increases, the critical field decreases.
- The relationship between the critical field and temperature can be described by the empirical formula: $H_c(T) = H_c(0)(1-(T/T_c)^2)$, where $H_c(T)$ is the critical field at temperature T , $H_c(0)$ is the critical field at absolute zero, and T_c is the critical temperature of the superconductor.
- The critical field also depends on the type of superconductor, with Type I superconductors having a lower critical field than Type II superconductors.

Persistent Current

- A persistent current is a steady electric current that flows indefinitely in a superconducting loop.
- This phenomenon occurs because superconductors have zero electrical resistance, allowing the current to flow without any energy loss.
- The current will continue to flow until an external force, such as a change in temperature or the application of a magnetic field, disrupts the superconducting state.
- Persistent currents are used in a variety of applications, including superconducting magnets and energy storage devices.
- In superconducting magnets, the persistent current generates a strong magnetic field that can be used for medical imaging, particle accelerators, and other applications.

- In energy storage devices, the persistent current can be used to store electrical energy with high efficiency and minimal energy loss.

Type I and Type II superconductors

- Type I superconductors have only one critical magnetic field. When that value is reached, the superconductive characteristics of a material are gone .
- Type II superconductors, on the other hand, have two critical magnetic fields .
- In type I superconductors, the transition from the normal state to the superconducting state occurs instantly, while in type II superconductors the transition happens slowly .
- By decreasing the temperature after the critical temperature, the superconducting properties of type II superconductors increase .
- In type II superconductors, some magnetic field lines can penetrate through the superconductor, while in type I superconductors it is impossible (i.e. magnetic field lines cannot penetrate through superconductors) .
- Most type II superconductors have a higher critical temperature .
- All high-temperature superconductors are type-II superconductors .
- While most elemental superconductors are type-I, niobium, vanadium, and technetium are elemental type-II superconductors .
- Boron-doped diamond and silicon are also type-II superconductors .
- Type-II superconductors are usually made of metal alloys or complex oxide ceramics .
- Type-I superconductors are generally pure metals .

High temperature superconductors

- High-temperature superconductors (abbreviated high-T_c or HTS) are defined as materials that behave as superconductors at temperatures above 77 K (−196.2 °C; −321.1 °F), the boiling point of liquid nitrogen.
- The main class of high-temperature superconductors is copper oxides combined with other metals, especially the Rare-earth barium copper oxides (REBCOs) such as Yttrium barium copper oxide (YBCO).
- The second class of high-temperature superconductors in the practical classification is the iron-based compounds.
- High-temperature superconductors, when discovered in 1986, presented a huge challenge but also opportunity to companies like ABB, GE, and Siemens to radically improve their products.
- A new experiment led by the condensed matter physicist Séamus Davis at the University of Oxford all but settles the origin of high-temperature superconductivity, a puzzle Davis has worked on for 25 years. When electrons couple up, further quantum trickery makes superconductivity unavoidable.

Properties and Applications of Super-conductors

Properties

- A superconductor is an element or metallic alloy which, when cooled below a certain threshold temperature, the material dramatically loses all electrical resistance.
- In principle, superconductors can allow electrical current to flow without any energy loss (although, in practice, an ideal superconductor is very hard to produce).
- Another important property of a superconducting material is its critical magnetic field $B_c(T)$, which is the maximum applied magnetic field at a temperature T that will allow a material to remain superconducting.

Applications

- Superconductors are used in a variety of applications, but most notably within the structure of the Large Hadron Collider. The tunnels that contain the beams of charged particles are surrounded by tubes containing powerful superconductors.
- Superconductors are used in particle accelerators, generators, transportation, computing, electric motors, medical, power transmission, etc.
- Superconductors are primarily employed for creating powerful electromagnets in MRI scanners. These conductors are used to transmit power for long distances.
- Most of the applications of the superconductor examples are because of their properties which provide advantages such as low power loss because of less dissipation of energy, high-speed operations because of zero resistance and continuous flowing electrical current, and high sensitivity.

Nano-Materials: Introduction and properties of nano materials

Nano-materials are materials with at least one dimension in the nanometer scale (1-100 nm). They exhibit unique properties due to their small size and high surface area to volume ratio. These properties can include enhanced mechanical, electrical, magnetic, optical, and chemical properties.

Some of the properties of nano-materials include:

1. High surface area to volume ratio: Due to their small size, nano-materials have a large surface area to volume ratio, which can enhance their reactivity and catalytic properties.
2. Quantum confinement: When the size of a material is reduced to the nanoscale, its electronic and optical properties can change due to quantum confinement effects.
3. Enhanced mechanical properties: Nano-materials can exhibit enhanced mechanical properties such as strength, toughness, and hardness due to

their small size and high surface area to volume ratio.

4. Enhanced optical properties: Nano-materials can exhibit enhanced optical properties such as absorption, scattering, and fluorescence due to their small size and high surface area to volume ratio.
5. Enhanced electrical and magnetic properties: Nano-materials can exhibit enhanced electrical and magnetic properties due to their small size and high surface area to volume ratio.

These unique properties of nano-materials make them useful in a wide range of applications, including electronics, energy, medicine, and materials science. They are also used in the development of new technologies such as nanoelectronics, nanophotonics, and nanomedicine.

Basics concept of Quantum Dots for the notes of the Unit 5 - Superconductors and Nano-Materials: in the subject of ENGINEERING PHYSICS

- Quantum dots (QDs) are colloidal fluorescent semiconductor nanocrystals, roughly spherical in shape .
- They are semiconductor particles a few nanometres in size .
- They have unique optical, electronic and photophysical properties that make them appealing in promising applications in biological labeling, imaging, and detection and as efficient fluorescence resonance energy transfer donors .
- Quantum dots are a central topic in nanotechnology and materials science .
- Quantum dots defined by lithographically patterned gate electrodes, or by etching on two-dimensional electron gases in semiconductor heterostructures can have lateral dimensions between 20 and 100 nm .
- Some quantum dots are small regions of one material buried in another with a larger band gap .
- Quantum dots are also brighter than a rival technology known as organic LEDs (OLEDs) and could potentially make OLED displays obsolete .

Quantum wires and Quantum well

Quantum wires and quantum wells are semiconductor structures that have applications in nanoelectronics, optoelectronics, information technology, and biotechnology, due to the quantum mechanical confinement provided by the small spatial extent of the structures.

- Quantum well materials are made of thin layers of differing semiconductor materials, and can be grown such that the movement of the free electron is confined to a two-dimensional plane.
- Quantum wires are one-dimensional structures where quantization occurs in two directions, leading to free movement along only one direction.

- Quantum wells and quantum well devices are a subfield of solid-state physics that is still extensively studied and researched today. The theory used to describe such systems uses important results from the fields of quantum physics, statistical physics, and electrodynamics.

Fabrication of Nano Materials - Top-Down Approach (CVD) and Bottom-Up Approach (Sol Gel)

Nano materials are materials with structures at the nanoscale, typically between 1 and 100 nanometers. There are two main approaches to fabricating nano materials: the top-down approach and the bottom-up approach.

Top-Down Approach (CVD)

The top-down approach involves breaking down larger materials into smaller nano-sized particles. One common method used in the top-down approach is Chemical Vapor Deposition (CVD). In CVD, a gas-phase precursor is introduced into a reaction chamber, where it decomposes and deposits onto a substrate to form a thin film of the desired material. The film can then be patterned and etched to create nanostructures.

Bottom-Up Approach (Sol Gel)

The bottom-up approach involves building nanostructures from smaller building blocks, such as atoms or molecules. One common method used in the bottom-up approach is the Sol Gel process. In the Sol Gel process, a solution of precursor molecules is prepared and then allowed to undergo a controlled chemical reaction to form a gel. The gel is then dried and heat-treated to form a solid material with nano-sized pores and particles.

Both the top-down and bottom-up approaches have their advantages and disadvantages. The top-down approach is well-established and can produce nanostructures with high precision, but it can be limited by the available precursor materials and the need for expensive equipment. The bottom-up approach allows for more control over the final nanostructure and can produce a wider range of materials, but it can be more difficult to control the size and shape of the final product.

In the field of engineering physics, the study of superconductors and nano-materials is an important topic. Understanding the fabrication methods for nano materials, such as the top-down and bottom-up approaches, is essential for the development of new materials and technologies.

Properties and Application of Nano Materials

Nanomaterials are materials with a particle size of less than 100 nm in at least one of its dimensions. They possess unique optical, magnetic, electrical, and other properties that emerge at this scale. These emergent properties have

great potential applications in various fields such as electronics, medicine, and more.

Some of the properties and applications of nanomaterials are:

1. **Optical properties:** The optical properties of nanomaterials are used to form optical detectors, sensors, lasers, displays, and solar cells. This property is also used in biomedicine and photoelectrochemistry. In microbial fuel cells, the electrodes are made up of carbon nanotubes.
2. **Unique material characteristics:** Due to their submicroscopic size, manufactured nanoparticles have unique material characteristics and may find practical applications in a variety of areas, including medicine, engineering, catalysis, and environmental remediation.
3. **Synthesis, properties, and applications:** Nanoparticles exist in different forms and can be classified into different classes based on their properties, shapes, or sizes. They have a size range from 1 to 100 nm.